A low-noise support bound on thirteen probability panels

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This independent check extends the direct low-noise argument to all thirteen closed panels in $2/5 \le p \le 53/100$. It uses no Fourier kernel, phase export or imported numerical program. The contact and Gaussian estimates below remain explicit analytic premises.

Precise conditional statement. Let X have mean zero and variance one, with an actual central gap $(-A, B)$, $A, B > 0$, and let $p = \mathbb{P}(X > 0)$, $q = 1 - p$. Write

$$m = \mathbb{E}X_+ = \mathbb{E}X_-, \quad \ell = m/(pq), \quad X = \ell(Z - p + U), \quad Z = \mathbf{1}_{\{X>0\}}, \quad \mathbb{E}(U \mid Z) = 0.$$

The sign of $Z - p + U$ agrees with that of $Z - p$. Put

$$\theta = \ell^2 \mathbb{E}U^2, \qquad \lambda = n\mathbb{E}U^2, \qquad \delta = 1 - (A + B)/\ell.$$

Assume $n \ge 42$, $0 \le \lambda \le 3/40$, $2/5 \le p \le 53/100$, the already acquired $|U| \le 3/5$, and $\delta < .072$. Assume the fixed-target closed-support contact inequality with a sharp predecessor and nonnegative excess, and its uniform Gaussian bounds specified below. These premises apply to the same law.

Theorem 1. *Under these hypotheses $|U| < 1/2$, including at every support atom. The conclusion holds for every raw threshold H and every actual $n \ge 42$. It makes no claim for larger noise or for other probability panels.*

Probability and moment bounds. Fix one closed interval $I = [l, u] = [i/100, (i + 1)/100]$, $i = 40, \dots, 52$, and write

$$v_- = \min_{p \in I} p(1 - p), \quad v_+ = \max_{p \in I} p(1 - p), \quad \Theta = \frac{3/40}{42v_- + 3/40}, \qquad E_I \le \sqrt{\frac{1 - \Theta}{v_+}}.$$

The maximum v_+ includes $1/4$ when the interval meets $1/2$. Variance gives $\ell^2(pq + \lambda/n) = 1$, hence $\theta = \lambda/(npq + \lambda) \le \Theta$ and $\ell \ge E_I > 0$.

Define $s_+ = \ell^2 \mathbb{E}[U^2 \mathbf{1}_{\{Z=1\}}]$ and $s_- = \ell^2 \mathbb{E}[U^2 \mathbf{1}_{\{Z=0\}}]$, so $s_+ + s_- = \theta$. Conditional centering and sign preservation give

$$\beta := \mathbb{E}[X|X|] = (1 - \theta)(1 - 2p) + s_+ - s_-.$$

Both endpoint functions decrease with p, including when $p > 1/2$. Therefore

$$b_- = 1 - 2u - 2(1 - u)\Theta \le \beta \le 1 - 2l + 2l\Theta = b_+. \tag{1}$$

There is no reflection of p to $\min(p, 1 - p)$.

Expand the absolute cubic using the same conditional marginal:

$$\rho := \mathbb{E}|X|^3 = \ell^3 \left\{ pq(1 - 2pq) + 3\mathbb{E}|Z - p|U^2 + \mathbb{E}[\operatorname{sgn}(Z - p)U^3] \right\}. \tag{2}$$

The linear term vanishes conditionally. Since $|Z - p| \le \max(p, q)$, $|U|^3 \le (3/5)U^2$ and $\ell^3 \le (pq)^{-3/2}$, the non-skeleton contribution is at most

$$\frac{3\max(p, q) + 3/5}{(pq)^{3/2}} \frac{\lambda}{n} \le 24\frac{\lambda}{n}.$$

This coefficient is valid on the whole range, since $pq \ge 6/25$, $\max(p, q) \le 3/5$, and exactly

$$24^2(6/25)^3 - (12/5)^2 = 34416/15625 > 0. \tag{3}$$

The function $(1-2v)/\sqrt{v}$ decreases for $0 < v \le 1/4$; the positive skeleton in (2) is multiplied by $(1-\theta)^{3/2} \le 1$. If $s_I \le \sqrt{v_-}$, this proves the usable same-law ceiling

$$\rho \le r_I := \frac{1-2v_-}{s_I} + \frac{24(3/40)}{42}. \tag{4}$$

No separate incompatible moment maxima are asserted to constitute a law.

Contact and whole-ray exclusion. The retained fixed-target contact inequality is

$$0 \le H_n(H,t) + c\{-|t|^3 + \tfrac{3}{2}\rho t^2 + 3\beta t + \rho e_n\} - na, \qquad t \in \operatorname{supp} P,$$

with $a \ge 0$, $c = C_{\mathrm{E}} \ge 819/2000$ and $e_n < 8/875$. The sharp predecessor is $C_{n-1} \le c$. Discarding $-na$ preserves an upper bound. For $t > \sqrt{3}$, set

$$S = 81/80, \quad A_t = (t^3 - 3t)/6, \quad H_4 = t^4 - 6t^2 + 3, \quad E_t = t^5/4100 + 3t/820 + 17(3t^2-2)/78720.$$

The retained uniform heat expansion and derivative estimate imply

$$H_n(H, \pm t) \le S^3\{\tfrac{2}{5}(A_t + |H_4|/72) + E_t\}, \qquad |\partial_t H_n(H, \pm t)| < \tfrac{2}{5}(t^2/2 + 1/2 + 8/875). \tag{5}$$

These are bounds for all real raw H. The absolute quartic term avoids a positive-H_4 assumption; $t^2 > 3$ ensures the cubic sign and the heat remainder's use of $|t^2 - 1| = t^2 - 1$.

For $r = r_I$, use $b = -b_-$ on the negative ray and $b = b_+$ on the positive ray, and define

$$Q(t) = -t^3 + \tfrac{3}{2}rt^2 + 3bt + (8/875)r, \qquad L(t) = \tfrac{2}{5}(t^2/2 + 1/2 + 8/875) + (819/2000)Q'(t).$$

At a boundary check

$$P(t) := S^3\{\tfrac{2}{5}(A_t + |H_4|/72) + E_t\} + (819/2000)Q(t) < 0, \tag{6}$$

and $Q, Q', Q'', L, L', L'' < 0$. The sign of Q licenses replacing c by its lower bound in (6). Since $Q''' = -6$ and $L'' = 2/5 - 6(819/2000) < 0$, these signs persist for all larger t. Equation (5), together with $Q' < 0$, then makes $H_n(H, \pm t) + cQ(t)$ strictly decreasing. Thus the entire exterior ray, including its boundary, contains no support point.

Take the rational boundaries

$$t_- = E_I(l + 1/2), \qquad t_+ = E_I(3/2 - u). \tag{7}$$

The independent exact calculation uses directed square roots of precision 10^{-12}, checks all thirteen closed intervals, and proves every boundary satisfies $t_\pm^2 > 3$. All 26 inequalities (6) and all 156 continuation signs pass strictly. The least favorable contact upper bound is on panel 40's positive ray and is less than $-.213$. Complete rational values are recorded in the accompanying JSON output; no sampled positivity is used.

Consequently $X \in (-t_-, t_+)$. Since $\ell \ge E_I$ and $p \in [l, u]$, this implies $Y = X/\ell + p \in (-1/2, 3/2)$. For $Z = 0$, the actual central gap gives $U = Y \le p - A/\ell = (\ell p - A)/\ell \le \delta < .072$. For $Z = 1$, it gives $U = Y - 1 \ge -(\ell q - B)/\ell \ge -\delta > -.072$. The exterior and inward bounds together yield $|U| < 1/2$. At $\lambda = 0$, conditional centering and zero conditional variance already give $U = 0$; no division by noise is used. The shared probability and noise endpoints are included.

Scope and verification. The earlier moment proof supplies $Y \in (-.6, 1.6)$ and $\delta \le .0719539227$ on panels 40–52; their inward consequences give precisely the prior $|U| < .6$ used above. Thus the new conclusion is not an input to its own proof. Seven negative controls reject, including an actual standardized law above $p = 1/2$ that refutes the shortcut $b_+ = 1 - 2l + \Theta$. The checker imports only the Python standard library. A numerical PASS verifies the recorded rational implications, while the contact, heat and acquired-strip premises retain their stated analytic status. No larger-noise confinement or finite-base result is claimed.

Continuous replacement of the thirteen new boundary rows. There is a shorter verification with common constants

$$\Theta_0 = 37/5000, \qquad E_0 = 249/125, \qquad r_0 = 221/200.$$

Indeed $5/677 < \Theta_0$, $E_0^2 < 4(1-\Theta_0)$, and $(r_0 - 3/70)^2(6/25) > (13/25)^2$, with $r_0 > 3/70$. These prove the requisite theta, scale and third-moment bounds for all $p \in [.4, .6]$, provided the same prior strip and inward bounds are supplied. The acquired entry application remains only $p \in [.4, .53]$.

For the negative ray set $z = p$; for the positive ray set $z = 1 - p$. These are physical support rays for the original law, with no CDF reflection. The pointwise signed beta bound is $b = 2z - 1 + 2(1 - z)\Theta_0$. At the boundary $t = E_0(z + 1/2)$, this is the affine function

$$b(t) = \frac{2(1 - \Theta_0)}{E_0} t - 2 + 3\Theta_0, \qquad t \in [t_0, t_1] = [1.7928, 2.1912].$$

The quartic H_4 is increasing on this interval and negative at t_1, hence $|H_4| = -H_4$. Substituting $b(t)$ into (6) therefore gives one explicit quintic $\widehat{P}(t)$, with leading coefficient $S^3/4100 > 0$.

The third derivative $\widehat{P}'''$ is a convex quadratic. The independent rational check gives strictly negative values at both endpoints, so $\widehat{P}''' < 0$ throughout. Consequently $\widehat{P}'$ is concave. Its two endpoint values are strictly positive; concavity puts it above their positive chord. Thus $\widehat{P}$ increases, and $\widehat{P}(t_1) < -.1779 < 0$ proves every boundary inequality. The proposed additional condition $\widehat{P}''(t_0) < 0$ is false: its value is approximately $+.0228525021$. It is not used in this corrected concavity argument; the failed preliminary check is preserved.

For the continuation gates distinguish the boundary-family polynomial $\widehat{Q}(t) = Q_{r_0, b(t)}(t)$ from derivatives holding beta fixed:

$$R_1(t) = -3t^2 + 3r_0 t + 3b(t), \qquad R_2(t) = -6t + 3r_0,$$

$$\widehat{L}(t) = \tfrac{2}{5}(t^2/2 + 1/2 + 8/875) + (819/2000)R_1(t), \quad L_1(t) = \tfrac{2}{5}t + (819/2000)R_2(t), \quad L_2 = \tfrac{2}{5} - 6(819/2000).$$

The exact gates $\widehat{Q}''(t_0) < 0$ and $\widehat{Q}'(t_1) > 0$ prove $\widehat{Q}$ increases, while $\widehat{Q}(t_1) < 0$. Both R_1 and $\widehat{L}$ have negative second derivatives and negative first derivatives at t_0, so they decrease from strictly negative values there. The affine functions R_2, L_1 also decrease from negative values, and $L_2 < 0$. All six fixed-beta radial signs therefore hold at every boundary. Their radial continuation was already justified above. This proves the same strict half-amplitude result without a new probability grid. The acquired thirteen-panel strip remains a separate antecedent; it has not been removed.